National University of Computer & Emerging Sciences

Chiniot-Faisalabad Campus

CL3001 – Machine Learning Lab (Spring 2026)

Project Paper Approval Request

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1. Selected Paper

Title: Performance Evaluation of Machine Learning and Deep Learning Models for Credit Risk Prediction

Authors: Irvine Mapfumo and Thokozani Shongwe

Venue & Date: *Journal of Risk and Financial Management* (MDPI), Vol. 19, Issue 3, March 2026

Paper Link: https://doi.org/10.3390/jrfm19030210

2. Datasets

Taiwanese Credit Card Default Dataset (UCI Repository):
https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients
30,000 client records with 24 financial and repayment features (22.1% default rate).

Statlog German Credit Dataset (UCI Repository):
https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data
1,000 loan applicant records with 20 credit and demographic features (30.0% default rate).

3. Reason for Choosing This Paper

I chose this paper because credit risk assessment is a practical problem in finance and the study compares core machine learning models covered in our course (Logistic Regression, Decision Tree, SVM, Random Forest, AdaBoost, XGBoost, and MLP) using 10-fold cross-validation. Both datasets are publicly available on the UCI repository and small enough to train and test thoroughly on standard hardware.

The paper also reports clear numerical baselines using SMOTE and SMOTE-ENN resampling (82.1% accuracy and 0.789 F1 on the Taiwanese dataset; 95.4% accuracy and 0.928 F1 on the German dataset). During replication, I plan to investigate how applying SMOTE-ENN before vs. inside cross-validation folds affects accuracy and probability calibration, and test whether probability calibration with cost-sensitive thresholding improves default risk estimation.

Instructor Approval: _____ Date: _____